

Benchtop Arbor Press

Model AP-1 · Mechanical assembly procedure

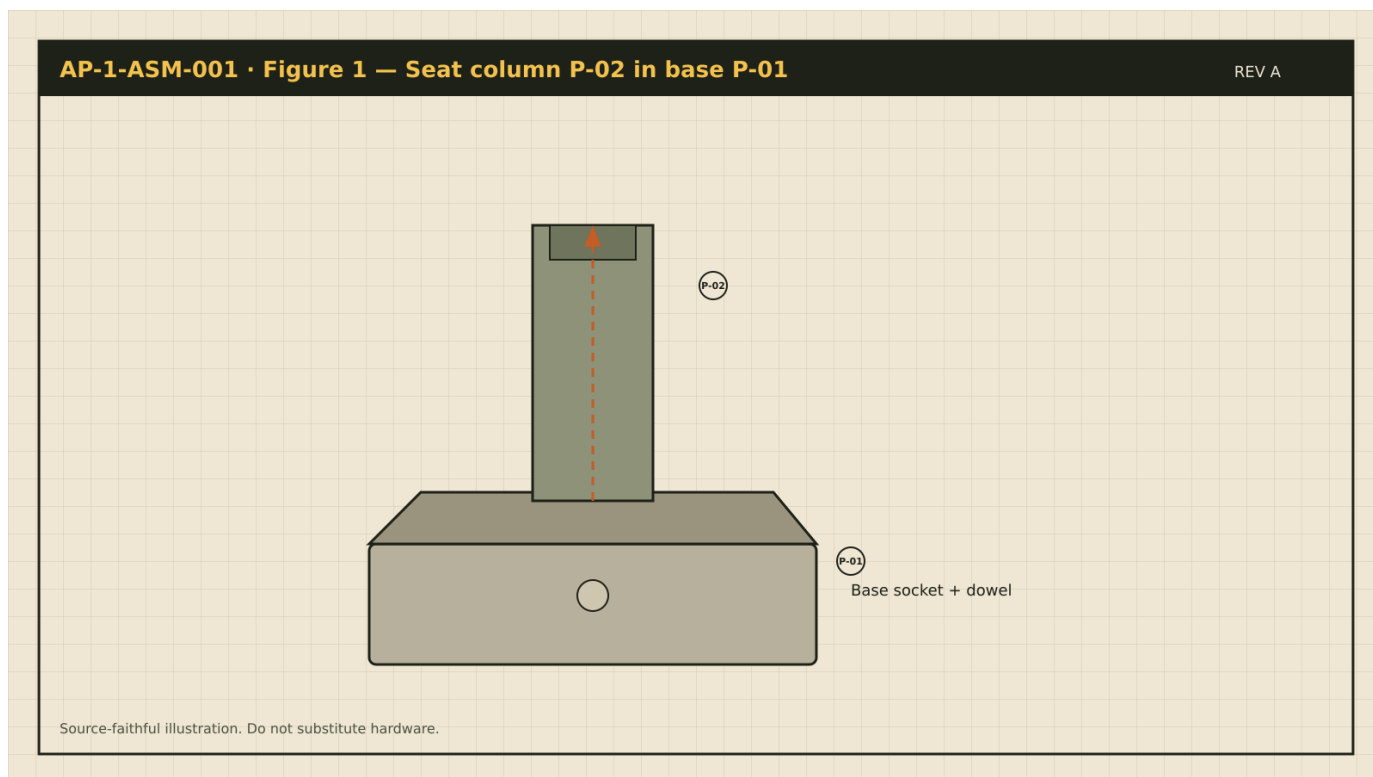
This kit manual is the source of truth for the AP-1 press.
Do not invent fasteners, tools, or torque values.

Tools: 4 mm hex key, 10 mm wrench, rubber mallet,
torque wrench 5–25 N·m.

Parts: P-01 base, P-02 column, P-03 ram, P-04 handle,
P-05 pinion shaft, P-06 rack guard, F-08 (×4),
F-10 (×2), W-10 (×2).

Step 1 — Seat the column in the base

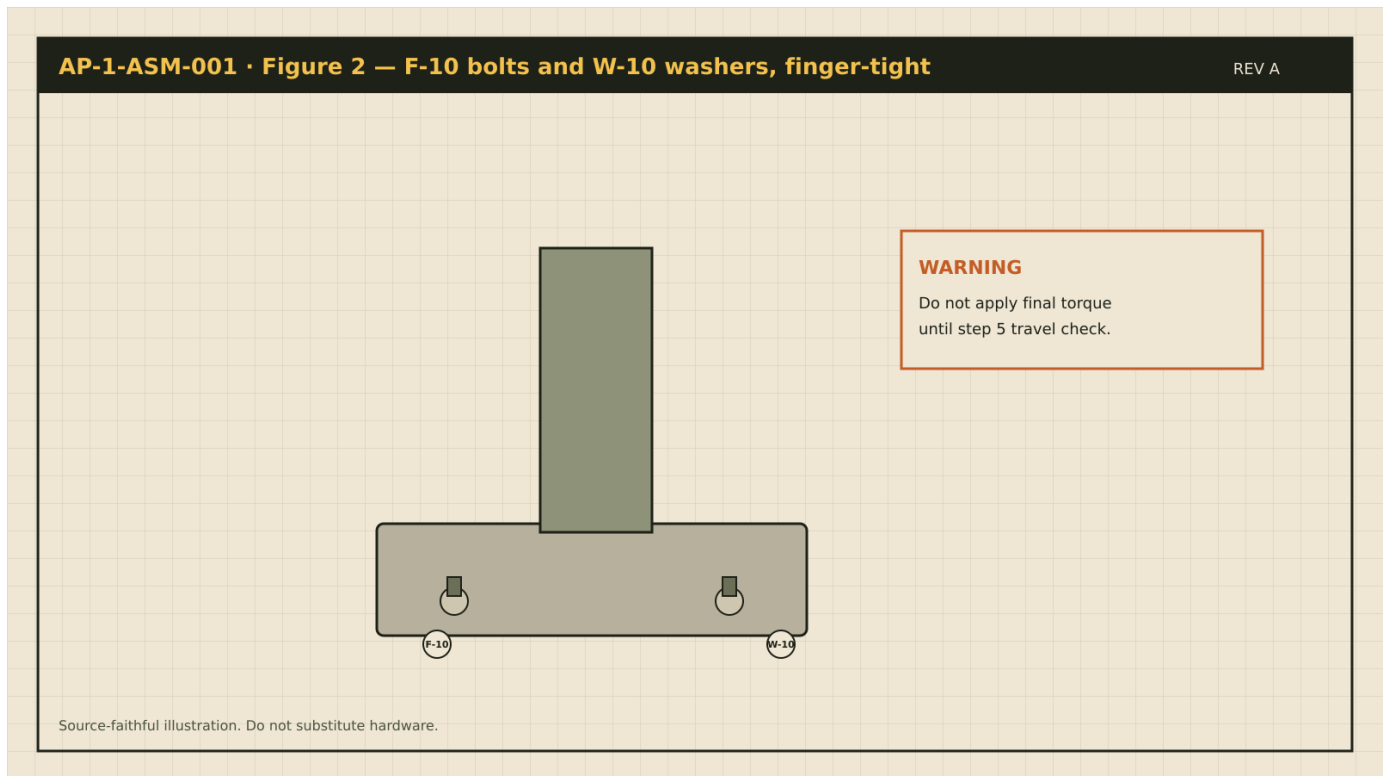
Stand base P-01 on a bench. Lower column P-02 into the base socket until the dowel engages. A rubber mallet may be used on a wood block. Do not force the fit.



Step 2 — Install base-to-column hardware

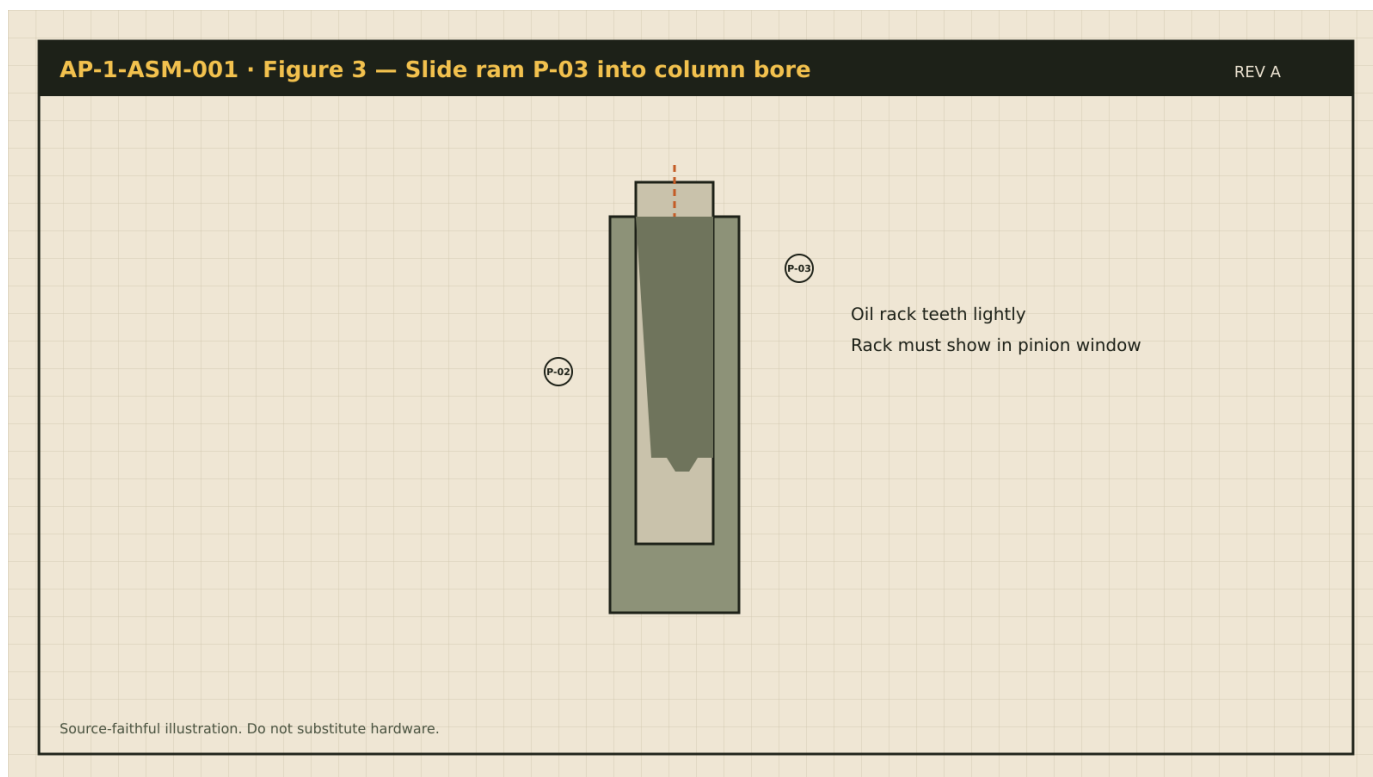
Pass two F-10 M10 × 30 hex bolts with W-10 lock washers up through the base into the column. Seat them finger-tight only.

WARNING: Do not apply final torque until ram travel is verified in step 5.



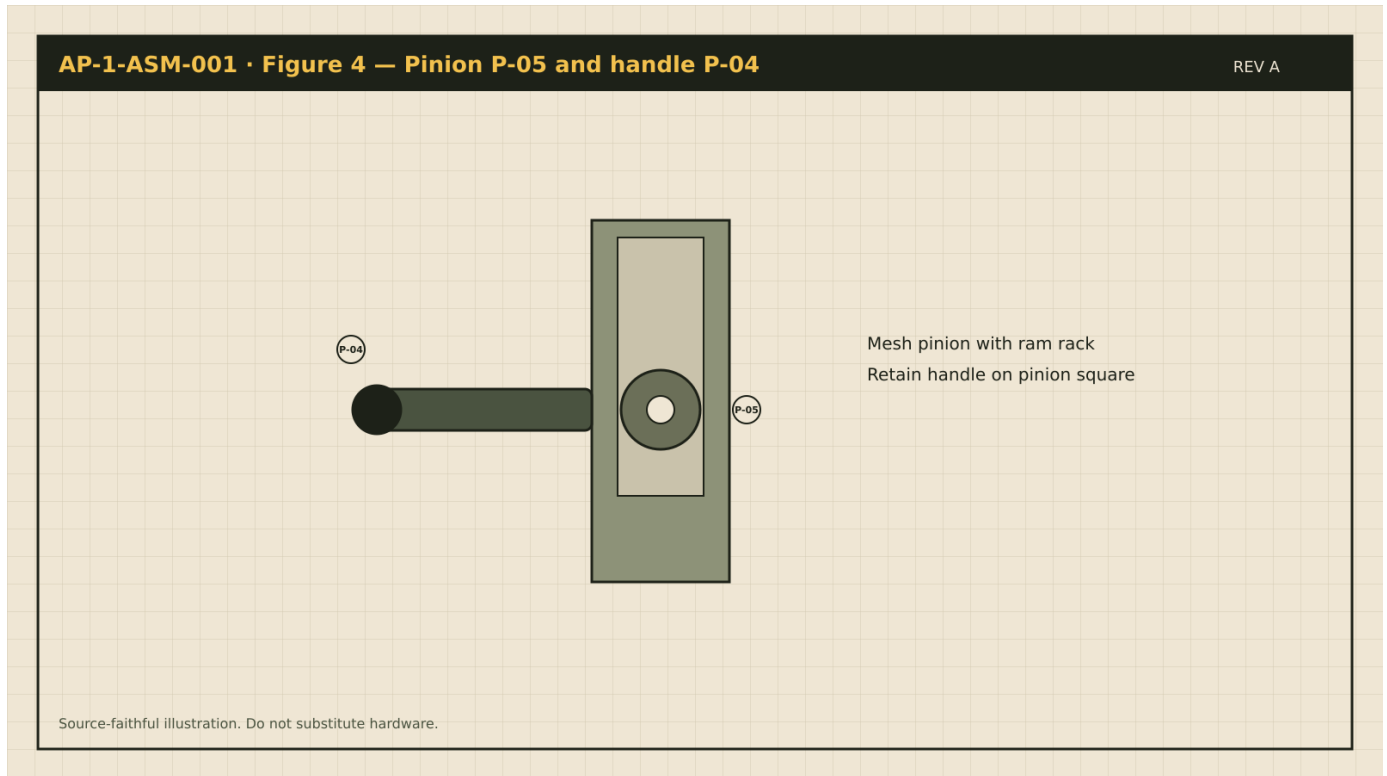
Step 3 — Slide the ram into the column

Wipe ram rack P-03. Apply a thin film of machine oil to the rack teeth. Slide the ram into the column bore from the top until the rack is visible in the pinion window.



Step 4 — Mesh the pinion and fit the handle

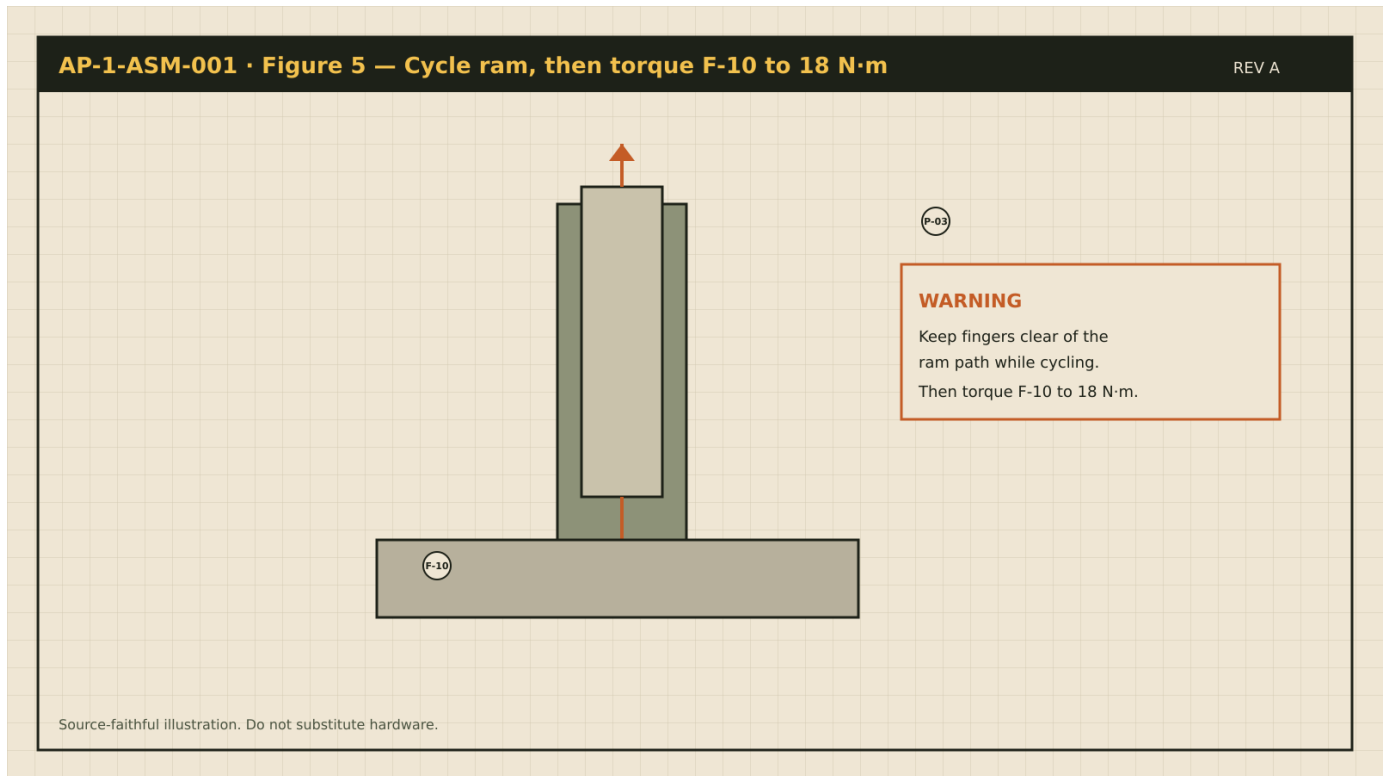
Insert pinion shaft P-05 through the column boss so the pinion teeth mesh with the ram rack. Fit handle P-04 onto the pinion square and retain it.



Step 5 — Verify travel, then torque the base

Cycle the ram through full travel. Confirm there is no bind. Then torque both F-10 bolts to 18 N·m.

WARNING: Keep fingers clear of the ram path while cycling.



Step 6 — Fit the rack guard

Place rack guard P-06 over the pinion window. Install four F-08 M8 × 25 socket cap screws and torque to 8 N·m.

